

Wavelet and Common Spatial Pattern for EEG Signal Feature Extraction and Classification

Liwei Zhang, Guozhong Liu, and Ying Wu

School of Photoelectric Information & Communication Engineering
Beijing Information Science & Technology University
Beijing, China
liweizhang2010@hotmail.com

Abstract—Brain-computer interface (BCI) can provide communication channels which do not depend on peripheral nerves and muscles for patients with neuromuscular disorders. The goal of the paper is to validate signal processing and classification methods for Brain-Computer Interfaces (BCIs). This paper presented a method combining wavelet with Common Spatial Pattern (CSP). Use multi-resolution analysis (MRA) to weaken noise and enhance features in the signals of motor imagery electroencephalogram (EEG). Then features are extracted and classification is completed by Common Spatial Pattern and Support Vector Machine (SVM) respectively. The classification accuracy achieves 93.5% in the course of testing on the data from subject. The result certifies the feasibility and effectiveness of this solution.

Keywords—brain-computer interface; wavelet transform; common spatial pattern; support vector machine

I. INTRODUCTION

Brain-Computer interface (BCI) can provide communication channels which do not depend on peripheral nerves and muscles for patients with neuromuscular disorders [1]. Sane patients suffering from neuromuscular disorders can directly control the prosthesis, light switches and other external things only by using motor imagery, which improves their self-care ability greatly. Therefore, the research on BCI has broad application prospects.

Limbs' action is most common in men's daily life. Studies have shown that when a unilateral limb (like the left hand) moves, the amplitudes of Mu rhythm (9-13 Hz) and Beta rhythm (18-22 Hz) of the brain contralateral sensorimotor cortex decrease markedly, while both the rhythms of ipsilateral sensorimotor cortex strengthen greatly. Moreover, when people only think of the action in his brain without real implementation, it will activate the same sensorimotor areas of the brain and produce the same EEG pattern as if the action were taken [2]. Therefore, motor imagery is an important paradigm of BCI in spontaneous EEG.

In this paper, three steps of signal preprocessing, feature extraction and pattern classification algorithm have been discussed. Denoising and feature enhancement are firstly implemented by wavelet transform. Then features are extracted and classification is completed by Common Spatial Pattern and Support Vector Machine (SVM) respectively.

II. METHODOLOGY

A. The algorithm of wavelet

The wavelet transform can be used to obtain the best representation of the underlying signal at different scales, up to a particular level [3].

In this paper, the Algorithm of MALLAT (Fig. 1) has been employed. Let $C_{j,k} = f_k$ then Multi-resolution signal analysis formula as follows

$$C_{j+1,m} = \sum_k \bar{h}(k-2m)C_{j,k} \quad d_{j+1,m} = \sum_k \bar{g}(k-2m)C_{j,k} \quad (1)$$

where $C_{i,k}$ is approximation coefficient, $d_{i,k}$ is details coefficient, so the corresponding reconstruction formula is

$$C_{j,k} = \sum_m h(k-2m)C_{j+1,m} + \sum_m g(k-2m)d_{j+1,m} \quad (2)$$

B. Common Spatial Pattern(CSP)

CSP algorithm is highly successful in calculating spatial filters for detecting ERD (event-related desynchronization) effects and for ERD-based BCIs [4]. The principle of the CSP algorithm is that the spatial component will gain maximum energy difference, i.e. maximize variance for one class and at the same time minimize variance for the other class [5].

Assume that X_1 and X_2 are the two classes of EEG signals evoked respectively by two mental tasks A and B [5]. The normalized spatial covariance matrices of the EEG can be obtained from

$$R_1 = X_1 X_1^T / \text{trace}(X_1 X_1^T) \quad R_2 = X_2 X_2^T / \text{trace}(X_2 X_2^T) \quad (3)$$

where $\text{trace}(M)$ is the sum of diagonal elements of matrix M and superscript T is transpose operator.

R can be factorized as follows:

$$R = R_1 + R_2 = U_0 \Sigma U_0^T \quad (4)$$

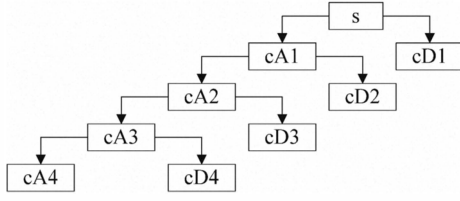


Figure 1. Sketch map of Mallat

where Σ is the diagonal matrix of eigenvalues and U_0 is the corresponding matrix of eigenvectors.

The whitening matrix is formed as

$$P = \Sigma^{-\frac{1}{2}} U_0^T \quad (5)$$

then,

$$Y_1 = PR_1P^T \quad Y_2 = PR_2P^T \quad (6)$$

Matrix R_1 and R_2 was decomposed as

$$Y_1 = U \Sigma_1 U^T \quad Y_2 = U \Sigma_2 U^T \quad (7)$$

Constructing spatial filter: Matrix U_1 and matrix U_2 were composed by the eigenvectors corresponding two largest eigenvalues. Hence, two spatial filters F_1 and F_2 can be defined as

$$F_1 = U_1^T P \quad F_2 = U_2^T P \quad (8)$$

The formula of source S is

$$S_1 = F_1 \times X \quad S_2 = F_2 \times X \quad (9)$$

where X is EEG signal of single trial.

C. Support Vector Machine(SVM)

A new machine learning method is Support Vector Machine which developed base on VC dimension theory and structural risk minimization principle in the statistical learning theory. It has advantages with strong adaptability, global optimization, high training efficiency, good generalization performance and small sample learning problem [6].

SVM is equivalent to solving a quadratic programming problem. Its goal is to find an optimal separating hyperplane (OSH) for a given feature set. A classifier implementing the optimal separating hyperplane in the feature space is given by,

$$f(x) = \text{sgn} \left[\sum_{i=1}^n y_i \alpha_i k(x, x_i) + e \right] \quad 0 \leq \alpha_i \leq C \quad (10)$$

where y_i is the labels of x_i , α_i is Lagrange multipliers, e is Classification threshold, $k(\cdot)$ is kernel function, x_i is support vectors, α_i and C are Lagrange multipliers and marginal factor respectively, which are significant to the performance of classifier. In this paper, the classification is carried out with Soft Margin Support Vector Machine whose kernel function is Radial Basis Function (RBF)

$$k(x, x_i) = \exp(-r \|x - x_i\|^2) \quad (11)$$

where r is width of RBF.

III. DATA PROCESSING

All the data used in experiments are from BCI Competition III dataset I. A subject had to perform imagined movements of either the left small finger or the tongue in experiments. The time series of the electrical brain activity was picked up during these trials using a 8x8 ECoG platinum electrode grid which was placed on the contralateral (right) motor cortex. All recordings were performed with a sampling rate of 1000Hz. Every trial consisted of either an imagined tongue or an imagined finger movement and was recorded for 3 seconds duration. 278 trials contained within training data and 100 in test data.

A. Data Preprocessing

Since large data set, we proceed down sampling and reduced the data dimension firstly. The sampling rate will be reduced to 200Hz, i.e. 600 dot every trail. We selected 10-lead data whose spectrum had obvious change as main data processing. Then the two EEG signals have filtered with 5-30Hz Butterworth band-pass filter separately.

B. Feature Extraction

Wavelet transform enhance the feature of EEG signals. Compiling program of soft-threshold de-noise and signal decomposition and reconstruction process, processed the 10-lead data. We adopt 4 levels of wavelet decomposition for discrimination, and reconstruct sub-band d3 and d4 to reduce noise.

Fig. 2 shows the original signal and wavelet-enhanced signal of two channels in one trial. As can be seen, the high frequency noisy signal has been effectively filtered and the detail components of signal was reserved after processing signal with the above method.

Feature extraction. The 278 trials in training data set was divided into two classes on category, that is, every class contain 139 trials. After having band pass filtered and having enhanced feature, we got two sets X_A and X_B whose size both were $600 \times 139 \times 10$. Two classes represent motor imagery of the left small finger and the tongue respectively. Each sample in every trials linked one by one to compose matrix X_1 and X_2 , the dimensions were 10×83400 . F_1 and F_2 were spatial filters which generated by CSP. After having spatial filtered the matrixs' dimensions were $2 \times 2 \times 600 \times 278$ (train data) and $2 \times 2 \times 600 \times 100$ (test data). Fig. 3 shows the signal has been filtered by CSP.

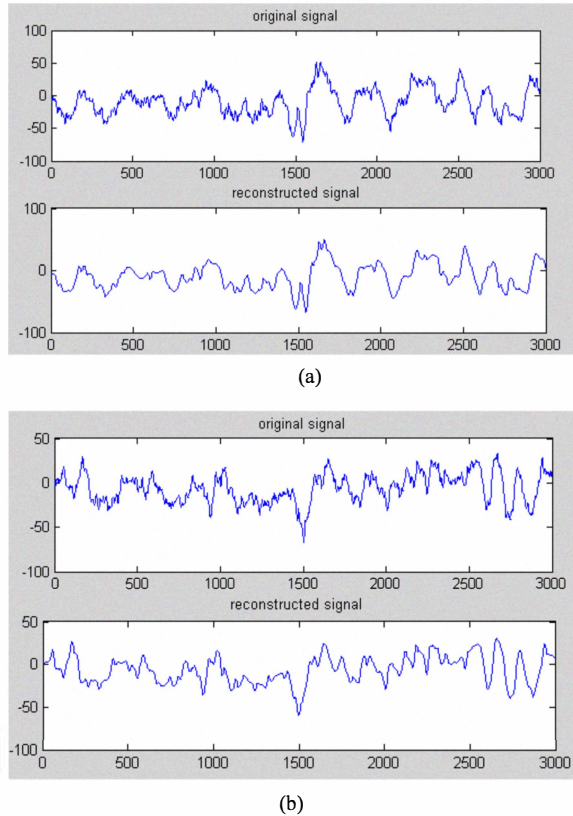


Figure 2. Two channels' signal enhanced by wavelet

IV. CLASSIFICATION RESULTS AND DISCUSSION

The above content discussed the method of extracting feature. The extracted eigenvector of training data will be taken as Support Vector Machine input. The kernel function of SVM was a Gaussian Radial Basis. Experiments proved that the optimal results will be got when the width of kernel function(r in formula (11)) was 1.2 in classification. Generally, the performance of SVM will be influenced by boundary coefficient C , the same results have received in diffent C because of algorithm Wavelet+CSP. The results was shown in Table I.

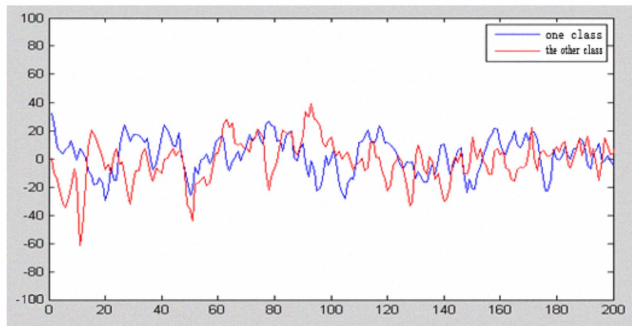


Figure 3. Signal filtered by CSP

TABLE I. DIFFERENT ACCURACY RATE WITH C

Penalty Coefficient C	CSP	Wavelet+CSP
10	91.7%	93.5%
100	91.8%	93.5%
150	91.8%	93.5%

An optimal wavelet basis, an exact wavelet decomposition level and a proper sub-band to be reconstructed are essential for excellent features enhancement. In view of signal's spectrum and the properties of wavelet decomposition, we adopt 4 levels of wavelet decomposition for discrimination. The wavelet basis of db4 is proved to be the optimal wavelet basis on minimum entropy criterion in experiments. This paper also involves threshold denoising based on SURE unbiased estimation, as well as reconstruction of sub-band d3 and d4 according to EEG signal's spectrum. Fig.2 describes a convinced effect of such an approach. CSP is employed to extract features for classification. In order to get best resolution, eigenvectors (corresponding with the largest eigenvalues in Σ_1 and Σ_2) are selected as the input vector of classifier.

V. CONCLUSION

The main contribution of this paper is signal feature extraction with wavelet transform and CSP filter. We take advantage of the terrific quality of wavelet transform on non-stationary signal processing to enhance signal features; then two class signal are processed by CSP filter to obtain maximum energy difference from each other. Thus the important information of the two classes has been kept and the noise has been removed. Wavelet+CSP has been proved to keep the stabilization of eigenvalue. The classification accuracy achieves 93.5% in the course of testing on the data from subject. The results of Table 1 show a fact that the compound method mentioned above is better than a single CSP solution on classification. The conclusion derived from Data set I is only an off-line analysis, however, further research on on-line analysis and action feedback are required for deep and comprehensive discussion on this subject.

ACKNOWLEDGMENT

The authors would like to thank Eberhard-Karls-Universität Tübingen for BCI data support and would like to express my heartfelt gratitude to my supervisor Dr. Guozhong Liu, whose keen and vigorous academic observation enlightens me not only in this thesis but also in my future study. The author is also grateful to my classmates for their valuable suggestions and the anonymous reviewers for their valuable comments.

REFERENCES

- [1] G. Schalk, J. Dennis, H. Thilo et al. "BCI2000: A General-Purpose Brain-Computer Interface (BCI) System," IEEE Trans. J. Biomed. Eng., vol. 51, no. 6, pp.1034-1043, 2004.
- [2] M. Jeannerod, "Mental imagery in the motor context," Neuropsychologia, J. Britain, vol. 33, no. 11, pp.1419-1432, 1995.

- [3] N. Hazarika, Jean Zhu Chen, A.C. Tsoi and A. Sergejew, "Classification of EEG signals using the wavelet transform," *Signal processing*, vol. 59, no. 1, pp.61~72, 1997.
- [4] S. Lemm, B. Blankertz, G. Curio, and K.-R. Müller, "Spatio-spectral Filters for Improved Classification of Single Trial EEG," *IEEE Trans. J. Biomed. Eng.*, vol. 52, no. 9, pp. 1541–1548, Sep. 2005.
- [5] Lin Liu, Qingguo Wei. "Extension of common spatial pattern (CSP) algorithm to multi-task case by Jacobi Rotations for single-trial EEG classification," *IEEE EMBS Conference on Neural Engineering*, J, Antalya, Turkey, pp.347-350, 2009.
- [6] S. Theodoridis, K. Koutroumbas, *Pattern recognition*(Third edition), China Machine Press, Beijing, China, 2006.